LINEAR ALGEBRA · LECTURE 1

# Systems of Linear Equations

Lecture Notes · Muhammad Zubair

## Learning outcomes

After this lecture you should be able to:

1. decide whether an equation is linear;
2. explain what a solution of a linear system is, and the terms *consistent* and *inconsistent*;
3. state the only three possible outcomes for a linear system and explain them with lines and planes;
4. write a system as $A\mathbf{x} = \mathbf{b}$ and as an augmented matrix $[A \mid \mathbf{b}]$;
5. use the three elementary row operations to solve a system, and read the answer from the last row.

## 1 Two stories to begin

### 1.1 A number game

Think of a number, double it, add 8, divide by 2, then subtract the number you started with. The result is always 4. Call the number $y$:

$$y \;\to\; 2y \;\to\; 2y+8 \;\to\; \frac{2y+8}{2} = y+4 \;\to\; (y+4)-y = 4.$$

We never needed to know $y$. **Algebra lets us reason about unknowns without knowing their values.** Every expression above is *linear* in $y$.

### 1.2 The Nine Chapters (China, about 200 BC)

The *Jiuzhang Suanshu* (Nine Chapters on the Mathematical Art) contains 246 practical problems. One of them:

> *Three bundles of grain A, two of B and one of C make 39 measures. Two of A, three of B and one of C make 34. One of A, two of B and three of C make 26. How many measures are in one bundle of each?*

With $x, y, z$ for the three classes:

$$\begin{aligned} 3x + 2y + z &= 39 \\ 2x + 3y + z &= 34 \\ x + 2y + 3z &= 26 \end{aligned} \qquad \Longrightarrow \qquad \left[\begin{array}{ccc|c} 3 & 2 & 1 & 39 \\ 2 & 3 & 1 & 34 \\ 1 & 2 & 3 & 26 \end{array}\right].$$

The ancient solvers wrote these numbers as a table on a counting board — what we now call a **matrix** — and eliminated entries step by step, exactly the method of this lecture. We solve it in Section 6.

## 2 Linear equations

**Linear equation (in my own words)**

Every variable appears only to the power 1, and no two variables are multiplied together. In general: $a_1x_1 + a_2x_2 + \cdots + a_nx_n = b$, where the $a_i$ and $b$ are constants.

**Example**

$x + 3y = 7$ is linear (its graph is a straight line). $x^2 + y = 4$ is **not** linear ($x$ is squared; the graph is a curve). $xy = 5$ is **not** linear ($x$ and $y$ are multiplied).

## 3 Systems of linear equations

A **system** is a collection of linear equations in the same unknowns. A **solution** is a list of numbers that makes *every* equation true at the same time. A system with at least one solution is **consistent**; with no solution it is **inconsistent**.

**Key idea**

Every linear system has exactly one solution, infinitely many solutions, or no solution. Nothing else is possible.

### 3.1 One system, three methods

Solve $x + y = 5,\ 2x - y = 4$.

- **Substitution:** $y = 5 - x$, so $2x - (5 - x) = 4$, $3x = 9$, $x = 3$, $y = 2$.
- **Elimination:** add the equations: $3x = 9$, so $x = 3$, $y = 2$.
- **Comparison:** $y = 5 - x$ and $y = 2x - 4$, so $5 - x = 2x - 4$, $x = 3$, $y = 2$.

Different roads, the same answer $(3, 2)$.

### 3.2 No solution and infinitely many solutions

**No solution**

$x + y = 4,\ 2x + 2y = 6$. Subtracting $2\times$(first) from the second gives $0 = -2$: impossible, so the system is inconsistent.

**Infinitely many solutions**

$x + 2y = 4,\ 2x + 4y = 8$. The same step gives $0 = 0$, always true. Let $y = t$ (any real number); then $x = 4 - 2t$. The solution set is $\{(4 - 2t,\ t) : t \in \mathbb{R}\}$.

### 3.3 Three equations, three unknowns

$$x + y + z = 6\ (1), \qquad x - y + z = 2\ (2), \qquad 2x + y - z = 1\ (3).$$

1. **Step 1** — (1) − (2)**.** The $x$ and $z$ terms cancel:

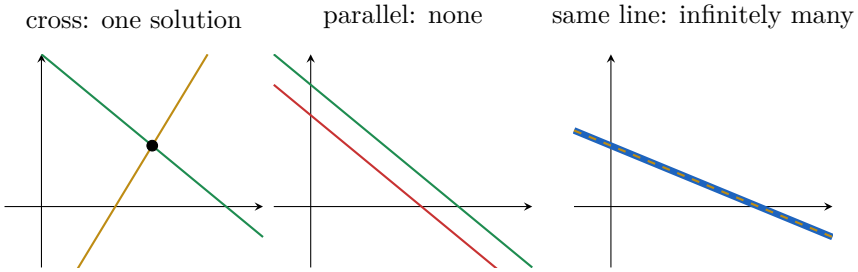
$$\begin{array}{rccccc} & x & + & y & + & z & = 6 \\ -( & x & - & y & + & z & = 2) \\ \hline & 0 & & 2y & & 0 & = 4 \end{array} \qquad \Rightarrow \qquad y = 2.$$

2. **Step 2 — put $y = 2$ into (1) and (3).** $x + 2 + z = 6 \Rightarrow x + z = 4, \quad 2x + 2 - z = 1 \Rightarrow 2x - z = -1.$
3. **Step 3 — add them.** The $z$ terms cancel: $3x = 3 \Rightarrow x = 1.$
4. **Step 4 — back-substitute.** $z = 4 - x = 4 - 1 = 3.$
5. **Check** in all three: $1 + 2 + 3 = 6$ ✓, $1 - 2 + 3 = 2$ ✓, $2 + 2 - 3 = 1$ ✓.

Solution $(x, y, z) = (1, 2, 3)$. Strategy: *three equations become two, two become one, then solve.* If equation (3) were replaced by $2x+2z = 5$, then (1)+(2) gives $2x+2z = 8$, contradicting $2x+2z = 5$ ($0 = 3$): no solution.

## 4 The geometry behind it

In two unknowns every linear equation is a straight line:




In three unknowns every equation is a **plane**. Three planes can meet at one point (one solution), share a whole line (infinitely many), or have no common point, for example a "tent" (no solution).

**Key idea**

The solution set is exactly where *all* the lines (or planes) meet.

## 5 Matrices and row operations

### 5.1 The computer's language: $A\mathbf{x} = \mathbf{b}$

$$\underbrace{\begin{bmatrix} 1 & 1 & 1 \\ 1 & -1 & 1 \\ 2 & 1 & -1 \end{bmatrix}}_{A} \underbrace{\begin{bmatrix} x \\ y \\ z \end{bmatrix}}_{\mathbf{x}} = \underbrace{\begin{bmatrix} 6 \\ 2 \\ 1 \end{bmatrix}}_{\mathbf{b}}, \qquad [A \mid \mathbf{b}] = \left[\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 1 & -1 & 1 & 2 \\ 2 & 1 & -1 & 1 \end{array}\right].$$

### 5.2 The three elementary row operations

1. **Swap** two rows: $R_i \leftrightarrow R_j$.

2. **Scale** a row by a nonzero number: $R_i \to cR_i$, $c \neq 0$.
3. **Replace** a row by itself plus a multiple of another row: $R_i \to R_i + cR_j$.

**Key idea**

These are exactly the moves we make with equations. They change how the matrix looks, never the solution.

### 5.3 Solving the 3-plane system

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 1 & -1 & 1 & 2 \\ 2 & 1 & -1 & 1 \end{array}\right] \xrightarrow[R_3-2R_1]{R_2-R_1} \left[\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 0 & -2 & 0 & -4 \\ 0 & -1 & -3 & -11 \end{array}\right] \xrightarrow{\cdots} \left[\begin{array}{ccc|c} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 2 \\ 0 & 0 & 1 & 3 \end{array}\right].$$

Full sequence (as in the video):

$$R_2 \to R_2 - R_1: \quad \left[\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 0 & -2 & 0 & -4 \\ 2 & 1 & -1 & 1 \end{array}\right]$$

$$R_3 \to R_3 - 2R_1: \quad \left[\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 0 & -2 & 0 & -4 \\ 0 & -1 & -3 & -11 \end{array}\right]$$

$$R_3 \to R_3 - \tfrac{1}{2}R_2: \quad \left[\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 0 & -2 & 0 & -4 \\ 0 & 0 & -3 & -9 \end{array}\right]$$

$$R_3 \to -\tfrac{1}{3}R_3, \quad R_1 \to R_1 - R_3: \quad \left[\begin{array}{ccc|c} 1 & 1 & 0 & 3 \\ 0 & -2 & 0 & -4 \\ 0 & 0 & 1 & 3 \end{array}\right]$$

$$R_2 \to -\tfrac{1}{2}R_2, \quad R_1 \to R_1 - R_2: \quad \left[\begin{array}{ccc|c} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 2 \\ 0 & 0 & 1 & 3 \end{array}\right]$$

Answer $x = 1,\ y = 2,\ z = 3$.

### 5.4 Read the last row

$\left[\begin{array}{cc|c} 1 & 2 & 3 \\ 0 & 0 & 5 \end{array}\right] \Rightarrow 0 = 5$: no solution; $\left[\begin{array}{cc|c} 1 & 2 & 3 \\ 0 & 0 & 0 \end{array}\right] \Rightarrow$ infinitely many; $\left[\begin{array}{cc|c} 1 & 0 & 1 \\ 0 & 1 & 2 \end{array}\right] \Rightarrow$ unique.

## 6 The grain problem, solved

Applying the same row operations:

$$\left[\begin{array}{ccc|c} 3 & 2 & 1 & 39 \\ 2 & 3 & 1 & 34 \\ 1 & 2 & 3 & 26 \end{array}\right] \longrightarrow \left[\begin{array}{ccc|c} 1 & 0 & 0 & 37/4 \\ 0 & 1 & 0 & 17/4 \\ 0 & 0 & 1 & 11/4 \end{array}\right].$$

So one bundle contains $9\frac{1}{4}$ measures of grain A, $4\frac{1}{4}$ of grain B and $2\frac{3}{4}$ of grain C. Check: $3(9.25) + 2(4.25) + 2.75 = 39$, $2(9.25) + 3(4.25) + 2.75 = 34$, $9.25 + 2(4.25) + 3(2.75) = 26$. ✓

## 7 Practice (Anton & Rorres, *Elementary Linear Algebra*, 11th ed., Exercise Set 1.1)

**9(a)** Is $(3, 1, 1)$ a solution of $2x_1 - 4x_2 - x_3 = 1,\ x_1 - 3x_2 + x_3 = 1,\ 3x_1 - 5x_2 - 3x_3 = 1$?
*Solution.* $6 - 4 - 1 = 1$, $3 - 3 + 1 = 1$, $9 - 5 - 3 = 1$. Yes.

**11(a)** $3x - 2y = 4,\ 6x - 4y = 9$. $R_2 - 2R_1$ gives $[\,0\ 0 \mid 1\,]$: no solution.

**11(b)** $2x - 4y = 1,\ 4x - 8y = 2$. $[\,0\ 0 \mid 0\,]$: infinitely many, $y = t$, $x = \frac{1}{2} + 2t$.

**11(c)** $x - 2y = 0,\ x - 4y = 8$. $R_2 - R_1$: $-2y = 8$, so $y = -4$, $x = -8$. Unique solution.

**19(a)** For which $k$ is $\left[\begin{smallmatrix} 1 & k & -4 \\ 4 & 8 & 2 \end{smallmatrix}\right]$ consistent? $R_2 - 4R_1$ gives $[\,0 \quad 8 - 4k \mid 18\,]$. If $k = 2$ this reads $0 = 18$. Consistent for all $k \neq 2$.

## 8 Summary

- A linear system is a set of linear equations that must all hold together.
- There are only three kinds of answers: one, infinitely many, or none.
- Lines (2D) and planes (3D) show *why*.
- $[A \mid \mathbf{b}]$ and the three row operations let a computer *find* the answer.

## Self-check

1. How many solutions does $x + y = 3,\ 2x + 2y = 6$ have? *(infinitely many — same line)*
2. What does a last row $[\,0\ 0 \mid 4\,]$ mean? *(no solution: $0 = 4$)*
3. Is $xy + z = 2$ linear? *(no — $x$ and $y$ are multiplied)*

**Next lecture:** Gaussian elimination.